Compression, Bypass, and Amplification: An Applied Stack-Audit Framework for Institutionally Connected Digital Assets

S. Meta

April 2026

Abstract

This paper extends the retained-demand framework developed in the preceding papers from conceptual analysis to applied stack-level evaluation. Earlier papers distinguished visible usage from retained demand, examined why settlement-stack expansion does not automatically close the need for external connective assets, and translated retained demand into conditional liquidity-density requirements. The present paper begins where that liquidity-density framework leaves off: when an institutional stack expands around a digital asset, does that stack increase the asset-specific liquidity-density burden, reduce it, route activity around it, or amplify the need to retain, position, mobilize, post, buffer, or maintain inventory in the asset itself?

The paper introduces a three-part stack-audit framework: compression, bypass, and amplification. Compression occurs when institutional efficiency, netting, internalized settlement, treasury optimization, custody aggregation, collateral efficiency, or substitute instruments reduce the need for asset-specific retained liquidity. Bypass occurs when the surrounding stack expands while the asset remains non-selected, weakly retained, or operationally replaceable. Amplification occurs only where stack design and actual usage increase the operational need to hold, mobilize, post, buffer, or maintain inventory in the asset itself.

Ripple- and XRP-adjacent institutional infrastructure is used as an applied stress environment rather than as proof of XRP necessity. The framework therefore rejects the assumption that Ripple's institutional expansion, RLUSD adoption, XRPL activity, custody growth, treasury tooling, tokenization initiatives, or collateral infrastructure automatically imply XRP-specific retained demand. Instead, each layer must be evaluated according to whether it reduces the asset-specific liquidity burden, routes activity around XRP, or increases the practical need for XRP-specific retained liquidity.

The paper emphasizes a removal test. If an asset can be removed from an institutional stack without materially worsening settlement efficiency, liquidity access, collateral mobility, timing, cost, capital efficiency, slippage, stress-time execution, or operational resilience, then ecosystem growth has not established strong retained-demand significance. Conversely, if removal creates measurable degradation in identifiable institutional contexts, an asset may possess meaningful connective value even without monopoly.

This is not a price thesis and does not propose a valuation target. Its contribution is methodological: it provides an applied audit structure for evaluating whether institutional stack expansion creates, weakens, redirects, or bypasses retained demand for an institutionally connected digital asset. The framework is designed to prevent both premature certainty and premature dismissal where infrastructure growth and asset-demand claims are easily conflated.

1. Introduction

The institutionalization of digital assets has created a persistent analytical ambiguity. As payment networks, stablecoins, custody platforms, tokenization systems, treasury tools, and settlement infrastructures expand, it becomes increasingly tempting to infer asset-level demand from ecosystem-level growth. Yet infrastructure expansion and asset-specific retained demand are not the same phenomenon. A digital asset may become more visible within an institutional environment without becoming more necessary to retain, position, mobilize, or buffer.

The preceding papers in this series developed a framework for separating these questions. The first paper distinguished visible usage from retained demand and argued that the relevant audit question is not merely whether an asset is used, but whether institutional environments create reasons to hold it in retained form. The second paper examined settlement-stack competition and argued that expansion does not necessarily imply closure: even where new rails, stablecoin systems, tokenized deposits, CBDC frameworks, or internalized settlement mechanisms expand, residual overflow and external connective burdens may persist under certain conditions. The third paper translated retained demand into conditional liquidity-density requirements, emphasizing usable float, available depth, turnover, slippage tolerance, and bounded market impact.

This paper asks the next applied question. Even if retained demand can be conceptually distinguished, even if residual overflow can persist, and even if required liquidity density can be conditionally sized, a further issue remains: what does a specific institutional stack do to the asset-specific liquidity burden? Does it compress that burden through efficiency and internalization? Does it bypass the asset by routing activity through substitutes or adjacent infrastructure? Or does it amplify the asset's retained-demand role by increasing the need to hold, mobilize, post, buffer, or maintain inventory in the asset itself?

In this sense, the present paper begins where the liquidity-density framework leaves off. If retained demand can be translated into required liquidity density, the next applied question is whether a specific institutional stack increases that burden, reduces it, redirects it, or avoids it altogether. Paper 4 does not replace the liquidity-density framework developed in Paper 3. It asks how actual institutional configurations move the burden identified there.[1]

This paper therefore forms the applied stack-outcome layer of the retained-demand audit series. Paper 1 defines retained demand by type, Paper 2 tests whether residual overflow survives settlement-stack closure, and Paper 3 translates retained demand into conditional liquidity-density requirements. The present paper asks whether actual institutional stack design compresses, bypasses, or amplifies that asset-specific burden. Later papers can then use this stack-outcome classification as an input for evidence-gated audit protocols and backend balance-sheet analysis.

The central claim of this paper is that stack expansion does not determine asset-specific retained demand by itself. Depending on design and actual usage, a specific institutional stack may compress, bypass, or amplify the retained-demand role of a connected digital asset. The appropriate task is therefore not to assume that ecosystem growth confirms asset necessity, nor to assume that new infrastructure eliminates it. The task is to audit the direction of the stack outcome.

The paper proceeds as follows. Section 2 translates the framework conditions developed in the preceding papers into applied stack outcomes. Sections 3 through 5 define compression, bypass, and amplification. Section 6 applies the framework to the Ripple- and XRP-adjacent institutional stack as a stress environment. Section 7 discusses derivatives, collateral, and float compression as conditional modifiers of

retained demand that can relocate the asset-specific burden without necessarily appearing in ordinary transactional use. Section 8 concludes.

2. From Framework Conditions to Stack Outcomes

The preceding papers established three analytical conditions that must be separated before any institutional claim about asset-level demand can be evaluated. The first condition is retained demand: whether the institutional environment creates reasons to retain the asset in held form rather than merely using it transiently. The second condition is residual overflow and selection: whether settlement-stack expansion leaves economically meaningful connective burdens that cannot be fully internalized, netted, or absorbed by substitute rails, and whether the relevant asset is actually selected for those burdens. The third condition is required liquidity density: whether the retained-demand role, if present, requires sufficient usable float, available depth, turnover, and slippage tolerance to support institutional-scale activity under bounded market-impact assumptions.

These conditions are necessary, but they are not yet sufficient for evaluating an actual institutional stack. A further applied step is required. Once retained demand, residual overflow, and liquidity-density requirements have been conceptually distinguished, the analysis must ask how a specific stack modifies the asset-specific burden. Institutional stack expansion may increase the scope of possible asset use, but it may also reduce the need to retain the asset, route activity through substitutes, or concentrate demand in ways that increase liquidity requirements. The relevant object of analysis is therefore not stack expansion by itself, but stack outcome.

This paper defines stack outcome as the direction in which a specific institutional configuration moves asset-specific retained demand and liquidity-density burden. Three outcomes are analytically distinct. First, a stack may compress the asset-specific burden by improving netting, internalization, treasury efficiency, prefunding reduction, collateral optimization, custody aggregation, or substitute settlement capacity. Second, a stack may bypass the asset by expanding adjacent infrastructure while routing activity through stablecoins, tokenized deposits, internal balance-sheet liquidity, omnibus arrangements, or other rails that do not require asset-specific retention. Third, a stack may amplify the asset's retained-demand role by increasing the need to hold, mobilize, post, buffer, or maintain inventory in the asset itself.

A stack outcome is therefore not a measure of ecosystem size. It is a directional classification of how a given institutional configuration changes the need to retain, mobilize, or buffer the asset itself. In this sense, stack-outcome analysis converts the liquidity-density question from a static requirement into an applied directional audit: does the stack increase the burden, reduce it, route around it, or concentrate it under specific institutional conditions?

The distinction is important because ecosystem-level growth can be consistent with all three outcomes. The same institutional expansion that creates more visible activity around a digital asset may reduce the need for retained balances, leave the asset operationally replaceable, or create deeper asset-specific inventory requirements. Without a stack-outcome framework, these possibilities are easily conflated. A growing ecosystem may be interpreted as evidence of asset demand even when the asset is being bypassed. Conversely, the presence of alternative rails may be interpreted as eliminating the asset's role even when residual overflow or stress-time liquidity requirements remain.

The purpose of the stack-audit framework is therefore to prevent premature inference. It does not assume that institutional expansion confirms asset necessity, nor does it assume that institutional efficiency

eliminates external connective assets. Instead, it evaluates whether a specific stack compresses, bypasses, or amplifies asset-specific retained demand under observable design and usage conditions.

Where evidence is insufficient to classify a layer as compression, bypass, or amplification, the appropriate classification is unresolved watch status. This status does not represent a fourth stack outcome, but a temporary audit state indicating that the available evidence does not yet justify directional classification.[4]

3. Compression: When Institutional Efficiency Reduces Asset-Specific Burden

Compression is the first stack outcome because it is the outcome most easily overlooked in asset-demand narratives. Institutional infrastructure can expand, professionalize, and become more efficient while reducing the need for asset-specific retained balances. In such cases, stack-level success may coexist with a weaker retained-demand burden for the connected asset.

Compression occurs when the institutional stack lowers the amount of asset-specific liquidity that must be held, positioned, or mobilized to support a given level of activity. This may occur through netting, internalized settlement, prefunding reduction, treasury optimization, omnibus custody arrangements, prime brokerage aggregation, liquidity-saving mechanisms, collateral substitution, or the use of stablecoins, tokenized deposits, and internal balance-sheet liquidity. Each mechanism can reduce the operational need to maintain retained inventory in the asset itself.

The key point is that compression should not be treated as failure at the infrastructure level. On the contrary, it may reflect institutional maturity. A payment system that reduces prefunding requirements, a treasury system that improves working-capital efficiency, a custody platform that aggregates balances, or a prime brokerage structure that nets exposures may all improve the efficiency of the surrounding stack. Yet those improvements may reduce the quantity of asset-specific retained liquidity required for settlement, collateral, or buffering purposes.

This distinction is especially important for institutionally connected digital assets. Many narratives treat institutional adoption as an automatic source of asset demand. However, institutional actors often prefer lower balance-sheet usage, reduced inventory requirements, minimized settlement exposure, and efficient collateral mobility. If a stack allows institutions to achieve the same or greater transaction capacity with fewer retained balances in the asset, then the stack has compressed the asset-specific liquidity burden.

Compression may take several forms. Settlement compression occurs when transactions are netted, internalized, or resolved through substitute settlement instruments before requiring movement through the asset itself. Treasury compression occurs when liquidity-management tools reduce the need for institutions to pre-position or retain the asset as working inventory. Custody compression occurs when omnibus structures, internal books, or prime-brokerage arrangements allow claims and balances to be managed without proportional on-ledger or asset-specific retention. Collateral compression occurs when exposure is reduced through netting, margin efficiency, rehypothecation, substitution, or acceptance of alternative collateral. Float compression occurs when the effective available supply for market use is narrowed or reorganized in ways that alter liquidity-density requirements, sometimes reducing day-to-day usage while increasing stress-time sensitivity.

Compression therefore has an ambiguous relationship to asset demand. In ordinary operating conditions, compression may reduce the need for retained balances. In stress conditions, however, the same compression may increase dependence on concentrated liquidity providers, high-quality collateral, or assets capable of rapid mobilization across venues. The audit question is therefore not merely whether

compression exists, but whether it reduces the asset's retained-demand role altogether or shifts it from routine transactional use to stress-time liquidity, collateral, or inventory functions.

A stack-audit approach must therefore ask four questions. First, which functions are being compressed: settlement, treasury, custody, collateral, or float? Second, does compression reduce the need to hold the asset, or merely reduce inefficient uses of it? Third, does compression substitute the asset with another instrument, or does it preserve the asset as a deeper liquidity backstop? Fourth, under stress conditions, does the compressed stack become less dependent on the asset, or more dependent on its ability to provide rapid liquidity, neutral settlement, or collateral mobility?

Compression is thus not a simple negative outcome. It is an efficiency outcome whose asset-demand implications depend on what is being compressed and what remains after compression. If compression removes the need to retain the asset, it weakens asset-specific retained demand. If compression eliminates only inefficient prefunding while preserving the asset as a stress-time bridge, collateral instrument, or liquidity buffer, it may transform rather than eliminate retained demand. The distinction must be evaluated at the level of actual stack design and usage, not inferred from ecosystem growth alone.

Compression and bypass should therefore be separated. Compression reduces the quantity, frequency, or intensity of asset-specific liquidity required within a stack. Bypass occurs when the relevant institutional function is performed without selecting the asset as a necessary component at all. The former may preserve the asset in a reduced or transformed role; the latter leaves the asset adjacent, optional, or replaceable.

Substitute instruments may produce compression when they reduce the amount of retained liquidity required within a still-relevant asset role; they produce bypass when they remove the asset from the relevant institutional function altogether.

4. Bypass: When Stack Expansion Does Not Produce Asset Retention

Bypass is the second stack outcome and the most important corrective to ecosystem-based demand inference. It occurs when the surrounding institutional stack expands while the asset itself remains non-selected, weakly retained, or operationally replaceable. In such cases, infrastructure growth may be real, adoption may be meaningful, and institutional use cases may expand, but the connected asset does not acquire a corresponding retained-demand role.

Importantly, bypass should not be treated as stack failure. It may occur precisely because the institutional stack has identified more efficient, compliant, liquid, or stable paths that do not require the asset in question.

Bypass differs from compression. Compression reduces the amount of asset-specific liquidity required to support activity. Bypass, by contrast, routes activity around the asset or allows institutional functions to develop without requiring the asset to be held, mobilized, posted, buffered, or maintained as inventory. Compression may still preserve the asset as a deeper liquidity layer or stress-time backstop. Bypass does not. It leaves the asset adjacent to the stack rather than functionally necessary within it.

This distinction is especially important in institutional digital-asset environments because stack expansion is often interpreted as asset validation. A payments network may grow without requiring the native or connected asset to be retained. A stablecoin may scale without using the asset as a bridge or liquidity buffer. A custody platform may support the asset without creating asset-specific inventory demand. A treasury platform may improve digital-asset operations generally without requiring a particular asset to be

held. A tokenization system may settle through stablecoins, tokenized deposits, internalized liquidity, or other rails without selecting the asset as a connective layer.

Bypass is therefore the applied form of infrastructure success without asset-specific retained demand. It is not evidence that the institutional stack has failed. On the contrary, the stack may be functioning well. The issue is that the asset's role remains peripheral, optional, or replaceable. For asset-demand analysis, this distinction is decisive. Institutional relevance at the stack level does not imply retained demand at the asset level.

Bypass may occur through several mechanisms. The first is substitution, where stablecoins, tokenized deposits, fiat rails, CBDC systems, or internal balance-sheet liquidity perform the relevant settlement or liquidity function. The second is internalization, where a platform, bank, broker, custodian, or treasury provider resolves obligations within its own ledger or controlled network without requiring asset-specific movement. The third is abstraction, where users interact with a service layer while the underlying asset becomes invisible, optional, or interchangeable. The fourth is compliance routing, where regulated institutions prefer instruments with clearer legal, accounting, or balance-sheet treatment. The fifth is incentive misalignment, where the operators of the stack benefit from ecosystem expansion but do not have economic reasons to create retained demand for the asset itself.

The most common error in bypass analysis is to treat adjacency as demand. An asset may be listed, supported, referenced, custodied, traded, bridged, or technically available without being economically necessary. Likewise, activity on a ledger or within an ecosystem may increase without creating reasons for institutions to retain the asset in meaningful size. The audit question is not whether the asset is present somewhere in the stack. The audit question is whether the stack requires the asset to be held or mobilized in a way that would be materially costly, inefficient, slower, riskier, or less resilient to remove.

This removal test is central. If an institutional stack can remove the asset without materially worsening settlement efficiency, liquidity access, collateral mobility, timing, cost, resilience, or stress-time execution, then the asset has not achieved an amplified retained-demand role. It may remain useful, tradable, visible, or strategically relevant, but it has not become necessary in the retained-demand sense developed in this series.

Bypass can also coexist with narrative amplification. A stack may generate more references to the asset, more public attention, more exchange listings, more integrations, or more speculative demand, while operational retained demand remains weak. This is why a stack-audit framework must distinguish narrative proximity from balance-sheet necessity. The relevant evidence is not rhetorical association but institutional behavior: retained balances, inventory requirements, collateral posting, liquidity-buffer positioning, market-making commitments, or operational dependency under stress.

For the Ripple- and XRP-adjacent context, bypass is not a marginal risk. It is one of the central outcomes to be audited. Ripple Payments may expand through fiat or stablecoin liquidity. RLUSD may scale as a dollar stablecoin without requiring XRP-mediated liquidity. XRPL activity may increase without translating into XRP retained balances. Ripple Custody may support institutional digital assets generally without creating XRP-specific inventory demand. Ripple Treasury may optimize enterprise digital-asset workflows without making XRP a necessary treasury asset. Tokenized assets may settle through stablecoins or internal rails. Each of these developments may be institutionally significant while still bypassing XRP retained demand.

Bypass therefore prevents overclaim. It allows the analysis to recognize institutional expansion without converting every expansion into an asset-demand claim. At the same time, it clarifies what would be required to move beyond bypass. The asset must become more than adjacent. It must become

operationally selected, retained, mobilized, or buffered in ways that are difficult to substitute without measurable degradation.

5. Amplification: When Stack Design Increases Asset-Specific Retained Demand

Amplification is the third stack outcome and the only one that strengthens the asset-specific retained-demand thesis. It occurs when institutional stack design and actual usage increase the operational need to retain, position, mobilize, post, buffer, or maintain inventory in the asset itself. Unlike ecosystem growth, visibility, technical support, or speculative attention, amplification requires evidence that the asset has become more necessary to the functioning of the stack.

Amplification is not triggered by mere adoption language. An asset is not amplified simply because it is mentioned in an institutional roadmap, supported by a custody platform, tradable on an exchange, or technically available for settlement. Nor is amplification established merely because a surrounding ecosystem grows. The relevant question is whether institutional actors acquire practical reasons to hold the asset in retained form, allocate balance-sheet capacity to it, maintain market-making inventory, use it as collateral, buffer liquidity through it, or rely on it under conditions where substitutes would materially worsen execution.

Amplification may arise through several channels. The first is settlement inventory. If institutions, liquidity providers, market makers, or payment intermediaries must maintain balances of the asset to support cross-venue, cross-currency, cross-ledger, or cross-institutional settlement, retained demand may increase. The second is liquidity-buffer demand. If the asset is held to absorb timing mismatches, bridge fragmented liquidity pools, or provide optionality during stress periods, the stack may create a buffer role. The third is collateral or margin recognition. If the asset becomes acceptable for margin, collateral, reserve, or security arrangements, and if that eligibility translates into actual posting or retained collateral balances, retained demand may be amplified. The fourth is market-making inventory. If institutions must hold the asset to provide reliable liquidity within acceptable slippage tolerances, the required liquidity-density burden may increase. The fifth is stress-time mobilization. If the asset becomes valuable because it can move liquidity across venues, jurisdictions, or settlement layers when ordinary routes are impaired, its retained-demand role may shift from routine usage to resilience infrastructure.

The strongest form of amplification occurs when removal of the asset produces measurable degradation. If removing the asset from the stack would materially increase settlement time, cost, slippage, prefunding needs, counterparty exposure, capital inefficiency, collateral friction, or stress-time execution risk, then the asset has moved beyond adjacency. It has become functionally embedded in the stack's liquidity architecture. This does not require monopoly. An asset may be one of several connective instruments and still have meaningful retained demand if its removal worsens execution in identifiable conditions.

This point is important because indispensability should not be defined as exclusive dominance. In multi-rail institutional environments, success may take the form of non-exclusive but material connective value. A digital asset can have an amplified retained-demand role if it serves as a preferred or efficient liquidity instrument for specific corridors, settlement windows, collateral contexts, market-making structures, or stress conditions. The relevant standard is not whether all activity depends on the asset, but whether a meaningful subset of institutional activity becomes materially less efficient without it.

Amplification must also be distinguished from speculative holding. Speculative demand may increase price exposure, trading volume, or market attention, but it does not necessarily indicate retained demand in the operational sense. Retained-demand amplification requires institutional reasons for holding that

arise from the design and usage of the stack itself. These reasons may include settlement readiness, liquidity provision, collateral eligibility, inventory obligations, or risk-management needs. The evidence should therefore be operational and balance-sheet oriented rather than purely market-sentiment oriented.

The relationship between amplification and liquidity density is direct. If a stack amplifies the asset's retained-demand role, the next question is whether the asset can support the corresponding liquidity-density burden. Increased institutional reliance may require sufficient usable float, reliable market depth, turnover, and acceptable slippage under both ordinary and stressed conditions. A stack may amplify demand in principle but fail in practice if liquidity is too fragmented, too shallow, too volatile, or too dependent on a small number of providers. Amplification therefore does not eliminate the need for liquidity-feasibility testing; it activates it.

For the Ripple- and XRP-adjacent context, amplification would require evidence that XRP is not merely present but operationally selected. Examples would include XRP being retained as settlement inventory by liquidity providers, used as a liquidity buffer across RLUSD or other tokenized-asset workflows, accepted and actually posted as collateral or margin, maintained by market makers to support institutional corridors, or relied upon as a stress-time bridge where removal would worsen cost, timing, slippage, or resilience. These are stronger claims than Ripple expansion, RLUSD issuance, XRPL activity, custody support, or treasury tooling by themselves.

Amplification is therefore the most demanding outcome in the stack-audit framework. It requires more than growth, more than visibility, and more than technical compatibility. It requires evidence that the institutional stack creates reasons to retain the asset in size, position it in advance, mobilize it under operational constraints, or preserve it as a liquidity or collateral option. Where such evidence is absent, the correct classification is not amplification, but compression, bypass, or unresolved watch status.

6. The Ripple Stack as an Applied Stress Environment

The Ripple- and XRP-adjacent institutional stack provides a useful applied stress environment for the framework developed in this paper. This is not because Ripple's institutional expansion proves XRP necessity. It does not. Rather, the Ripple context is analytically useful because it contains multiple layers in which ecosystem growth, infrastructure adoption, asset support, liquidity tools, stablecoin issuance, custody services, treasury functions, and tokenization initiatives may be easily conflated with XRP-specific retained demand.[2]

The following discussion is illustrative rather than exhaustive. It does not attempt a complete empirical assessment of Ripple's institutional stack. Its purpose is to show how the compression-bypass-amplification framework can be applied across layers where asset-demand inference is especially prone to overreading. It should therefore be read as an applied classification exercise rather than as a factual finding that any particular Ripple-related layer has already produced compression, bypass, or amplification.

For this reason, the Ripple stack should not be treated as a confirmation case. It should be treated as a stress environment for distinguishing compression, bypass, and amplification. Each layer of the stack may move the asset-specific retained-demand burden in a different direction. Some layers may reduce the need for XRP-specific retained balances. Some may expand institutional activity while routing around XRP. Others may create conditions under which XRP becomes more relevant as settlement inventory, liquidity buffer, collateral instrument, market-making asset, or stress-time bridge. The analytical task is to classify these effects rather than assume them.

Ripple Payments illustrates the ambiguity. Payment-network expansion may increase the number of corridors, counterparties, settlement use cases, and institutional relationships connected to Ripple infrastructure. However, this does not automatically imply XRP retained demand. If payment flows are resolved through fiat liquidity, stablecoins, internalized balance-sheet liquidity, prefunding optimization, or other non-XRP mechanisms, the result may be compression or bypass. XRP-specific amplification would require evidence that institutions or liquidity providers must retain XRP inventory, use XRP to bridge fragmented liquidity, or rely on XRP under conditions where alternative routes materially worsen cost, timing, slippage, or resilience.

RLUSD introduces a second ambiguity. A regulated dollar stablecoin may expand Ripple's institutional reach and settlement capacity. Yet stablecoin growth can compress or bypass XRP demand if RLUSD itself becomes the preferred settlement, treasury, or liquidity instrument. RLUSD adoption therefore cannot be treated as XRP demand by default. The relevant audit question is whether RLUSD increases the need for XRP-mediated liquidity, XRP-based market-making inventory, cross-asset buffers, or stress-time settlement optionality. If RLUSD scales independently of XRP, the classification may be bypass. If RLUSD reduces the need for XRP balances, the classification may be compression. If RLUSD creates new cross-ledger or liquidity-buffer roles in which XRP must be retained or mobilized, amplification may become plausible.

XRPL activity creates a third ambiguity. Ledger activity, token issuance, decentralized exchange use, automated market-making, or tokenization on XRPL may increase XRP's visibility and utility. However, ledger activity does not necessarily equal XRP retained demand. If XRP is used only transiently for fees or technical settlement while institutional retained balances remain minimal, the asset-demand implication is limited. Amplification would require evidence that XRP itself becomes necessary as inventory, liquidity bridge, collateral, buffer, or settlement medium in institutional size.

Ripple Custody and related institutional custody services create another classification problem. Custody support may make it easier for institutions to hold XRP, but support is not the same as necessity. A custodian may support XRP alongside many digital assets without creating XRP-specific inventory demand. Custody can even enable bypass if institutions hold tokenized assets, stablecoins, or other instruments through the same infrastructure while XRP remains optional. Amplification would require evidence that custody services support actual retained XRP balances for settlement readiness, collateral posting, liquidity buffering, market-making, or treasury allocation.

Ripple Prime and brokerage-like functions may also compress, bypass, or amplify. Prime structures can compress liquidity burdens through netting, aggregation, internal books, and efficient collateral management. They can bypass XRP if institutional execution and financing occur through other assets or stablecoin liquidity. They can amplify XRP if market makers, counterparties, or institutional clients must maintain XRP inventory to provide reliable execution, cross-venue liquidity, or collateralized exposure within acceptable slippage and settlement constraints.

Ripple Treasury provides a further example of the need for careful classification. Treasury tooling may expand the institutional use of digital assets by improving cash management, liquidity operations, reconciliation, reporting, and operational control. Yet treasury efficiency may reduce rather than increase asset-specific retained balances. If enterprises use treasury tools to hold stablecoins, tokenized cash, or other digital assets without retaining XRP, the outcome is bypass. If treasury optimization reduces idle XRP balances, the outcome may be compression. If treasury workflows create reasons to retain XRP as liquidity buffer, cross-rail settlement inventory, or stress-time optionality, amplification may be considered, but only under observable usage conditions.

Tokenization and RWA settlement also require non-automatic classification. Tokenized assets may settle through stablecoins, tokenized deposits, internal ledgers, or bank-controlled rails without selecting XRP. In such cases, tokenization expands the institutional field but may bypass XRP. Alternatively, tokenized markets may create fragmented liquidity pools, settlement windows, collateral frictions, and cross-ledger mobility problems that increase demand for neutral connective assets. XRP-specific amplification would require evidence that XRP is selected to address these frictions in a retained, mobilized, or buffered form.
The applied audit therefore proceeds layer by layer. For each Ripple- or XRP-adjacent function, the relevant questions are: does this layer reduce the need for XRP-specific retained liquidity; does it route activity around XRP; or does it create stronger operational reasons to retain, position, mobilize, post, buffer, or maintain inventory in XRP? Only the third pathway supports amplification. The first two pathways may still indicate institutional progress, but they do not support XRP-specific retained demand.
This approach also clarifies the evidentiary standard. Public announcements, integrations, technical compatibility, custody support, exchange access, or ecosystem activity are not sufficient by themselves. Stronger evidence would include retained XRP balances by institutional actors, market-maker inventory requirements, XRP-based liquidity buffers, actual collateral posting, settlement workflows where XRP removal causes measurable degradation, or stress-time dependence on XRP liquidity. Without such evidence, the appropriate classification remains compression, bypass, or unresolved watch status rather than amplification.
The Ripple stack is therefore valuable for this paper precisely because it resists simple interpretation. It contains plausible pathways to all three outcomes. A weak analysis would convert every Ripple development into XRP demand, or conversely treat every stablecoin or treasury development as eliminating XRP relevance. A stack-audit analysis does neither. It asks which layer is expanding, what function is being served, what asset is being retained, what liquidity burden is being shifted, and whether XRP becomes more necessary, less necessary, or merely adjacent.

7. Derivatives, Collateral, and Float Compression

Derivatives, collateral, and float compression form a secondary but important application layer for the stack-audit framework. They are treated separately because they can alter retained demand without necessarily increasing ordinary transactional use. In institutional markets, asset demand may arise not only from payment settlement but also from margin, collateral, hedging, inventory, liquidity buffering, and stress-time mobilization. These functions can amplify retained demand, but they can also compress or bypass it depending on market design.
This section is not intended as a full theory of derivatives, collateral markets, or float dynamics. It treats these domains as conditional modifiers of stack outcomes because they can relocate retained demand from visible transactional use toward inventory, margin, hedging, collateral, and stress-time liquidity. Its role is limited to showing how these domains can modify the compression-bypass-amplification classification rather than to provide a standalone account of financialization.
Derivatives can increase asset-specific retained demand when market participants must hold or access the asset to hedge exposure, provide liquidity, meet settlement obligations, post margin, or manage basis risk. A derivatives market may require market makers to maintain inventory, liquidity providers to hedge dynamically, or counterparties to access the underlying asset during settlement or liquidation events. In such cases, derivatives activity may amplify the asset's liquidity-density burden even if spot transactional use remains limited.

However, derivatives can also compress direct retained demand. If exposure can be netted, cash-settled, centrally cleared, or collateralized with substitute assets, the need to hold the underlying asset may be reduced. Derivatives may increase notional activity without increasing retained balances in the asset itself. This distinction is essential. Notional exposure, open interest, or trading volume should not be treated as retained demand unless they translate into inventory, collateral, hedging, settlement, or stress-time liquidity requirements involving the asset.

Collateral eligibility presents a similar ambiguity. If an asset becomes eligible collateral but is rarely posted, the retained-demand effect is weak. If eligibility leads institutions to retain balances in order to post margin, secure financing, support settlement, or manage liquidity under stress, the effect may be stronger. The relevant evidence is not eligibility alone, but actual collateral use, haircuts, concentration limits, liquidity treatment, margin practices, and the cost of substituting other collateral.

Collateral may also create bypass. Institutions may prefer cash, Treasury securities, tokenized deposits, stablecoins, or other digital collateral with clearer regulatory, accounting, liquidity, or risk-management treatment. In such cases, collateral infrastructure may expand while the asset remains unused or weakly used. The result is not amplification but bypass. The same collateral stack that could have supported asset-specific retained demand may instead route around it.

Float compression adds another layer of complexity. Available float is not identical to total supply. A large portion of supply may be held long term, locked, illiquid, strategically retained, operationally unavailable, or unwilling to move at current market conditions. If institutional demand begins to rely on the asset while usable float remains limited, the liquidity-density burden may increase sharply. Conversely, if stack design reduces the need for routine asset movement, visible usage may decline even while the asset's stress-time role becomes more important. This reorganization of float directly alters the required liquidity-density burden identified in Paper 3: it may lower routine requirements while increasing sensitivity to available depth during stress events.

Float compression can therefore support different outcomes. It may weaken ordinary usage demand if fewer balances are needed for day-to-day activity. It may amplify liquidity-density requirements if the asset is required under stress but only a limited share of supply is available. It may also create fragility if institutional reliance grows faster than usable float, market depth, or turnover. In this sense, float compression is not inherently bullish or bearish. It is a condition that changes the sensitivity of retained demand to liquidity availability.

The interaction between derivatives, collateral, and float compression is especially important for institutionally connected digital assets. A market may appear liquid during ordinary conditions because derivatives and internalized settlement reduce the need for spot movement. But during stress, margin calls, collateral substitution, liquidity withdrawals, or cross-venue settlement needs may suddenly require access to the underlying asset. If usable float is limited, stress-time liquidity may become the binding constraint.

For a stack-audit framework, the key questions are therefore concrete. Do derivatives markets require actual asset inventory, or do they merely create synthetic exposure? Does collateral eligibility translate into actual posted balances? Are market makers required to retain the asset to manage hedging, settlement, or liquidation risk? Does float compression reduce routine retained demand or increase stress-time liquidity sensitivity? Are substitute assets sufficient under ordinary and stressed conditions? Does removal of the asset from the derivatives or collateral stack materially worsen liquidity, cost, risk management, or settlement resilience?

This section does not argue that derivatives or collateral automatically amplify retained demand. It argues that they can shift the location of retained demand from payments and visible usage toward inventory, hedging, margin, collateral, and stress-time liquidity. Where those functions require the asset itself, amplification may occur. Where those functions are cash-settled, netted, substituted, or internalized, compression or bypass may be the more accurate classification.

The implication is that derivatives, collateral, and float compression should be treated as conditional modifiers rather than standalone proof of asset necessity. They can deepen asset-specific retained demand, but only where actual market structure requires the asset to be retained, posted, mobilized, or accessed under binding liquidity constraints. Without such evidence, increased financialization may produce more activity around the asset without producing stronger retained demand for the asset itself.

8. Conclusion

This paper has extended the retained-demand framework from conceptual analysis to applied stack-level evaluation. The central claim is that institutional stack expansion does not determine asset-specific retained demand by itself. A growing institutional environment may reduce the need for asset-specific retained liquidity, route activity around the asset, or increase the need to retain, position, mobilize, post, buffer, or maintain inventory in the asset itself. These outcomes must be distinguished rather than inferred from ecosystem growth.

The framework introduced in this paper classifies stack outcomes into three categories: compression, bypass, and amplification. Compression occurs when institutional efficiency, netting, internalized settlement, treasury optimization, custody aggregation, collateral efficiency, or substitute instruments reduce the asset-specific liquidity burden. Bypass occurs when the surrounding stack expands while the asset remains non-selected, weakly retained, or operationally replaceable. Amplification occurs only where stack design and actual usage increase the operational need to retain, mobilize, post, buffer, or inventory the asset itself.

This classification matters because institutional expansion can be interpreted too quickly in both directions. A favorable overreading converts every payment integration, stablecoin launch, custody service, treasury tool, or tokenization initiative into evidence of asset demand. A dismissive overreading assumes that any new stablecoin, internalized rail, treasury system, or collateral substitute eliminates the role of external connective assets. Both interpretations are premature. The relevant question is not whether the ecosystem is growing, but how that growth changes the asset-specific retained-demand burden.

Ripple- and XRP-adjacent infrastructure illustrates why such a framework is necessary. Ripple Payments, RLUSD, XRPL, Ripple Custody, Ripple Prime, Ripple Treasury, tokenization, collateral tooling, and institutional liquidity services may all be important developments. However, their implications for XRP-specific retained demand are not automatic. Each layer may compress, bypass, or amplify XRP's role depending on actual design, institutional behavior, balance-sheet treatment, liquidity provisioning, collateral use, and stress-time dependence.

The paper therefore does not conclude that XRP must become indispensable. It also does not conclude that XRP becomes irrelevant whenever adjacent infrastructure expands. Its contribution is to provide an applied audit structure for deciding which interpretation is warranted. Evidence of amplification would require more than visibility, compatibility, or narrative proximity. It would require observable reasons for institutions, liquidity providers, market makers, or counterparties to retain XRP, maintain XRP inventory,

use XRP as collateral or buffer, mobilize XRP across settlement contexts, or depend on XRP where removal would materially worsen timing, cost, slippage, liquidity access, capital efficiency, or resilience. The removal test is therefore central. If an asset can be removed from an institutional stack without materially degrading settlement efficiency, liquidity access, collateral mobility, timing, cost, stress-time execution, or operational resilience, then the asset has not achieved strong retained-demand significance. If removal creates measurable degradation in specific institutional contexts, the asset may possess a meaningful retained-demand role even without monopoly. In multi-rail systems, indispensability should not be defined as exclusive dominance, but as material connective value under identifiable conditions.

Derivatives, collateral, and float compression reinforce this point. Financialization may increase activity around an asset without increasing retained demand for the asset itself. Notional exposure, collateral eligibility, or trading volume should not be confused with retained balances, inventory requirements, actual posting, hedging needs, or stress-time liquidity access. At the same time, if derivatives, collateral markets, or compressed float create binding needs to access the asset under institutional constraints, they may shift retained demand from visible transactional use toward inventory, margin, collateral, or resilience functions.

The broader implication is methodological. For institutionally connected digital assets, the central audit question is not whether the asset appears within a growing ecosystem. The stronger question is whether the ecosystem creates reasons to retain the asset in held form, whether substitute rails close or bypass its role, whether liquidity-density requirements are feasible, and whether specific stack design compresses, bypasses, or amplifies the asset-specific burden.

Paper 4 therefore completes the applied stack-outcome layer of the retained-demand audit series by moving from retained-demand theory to stack-outcome classification. It does not offer a price target, a deterministic adoption claim, or a promotional thesis. It offers a disciplined way to evaluate whether institutional infrastructure growth creates asset-specific retained demand, reduces it, or routes around it.

Within the broader series, this paper functions as the transition point between conditional liquidity-density analysis and later evidence-gated operational audit: it identifies whether stack growth compresses, bypasses, or amplifies the asset-specific burden that subsequent evidence protocols must test.

Future work may apply the framework to specific institutional datasets, collateral-market structures, derivatives markets, or individual stack configurations in greater empirical detail. Such work would extend the present paper's applied audit structure without changing its core claim: institutional infrastructure growth must be evaluated according to whether it compresses, bypasses, or amplifies asset-specific retained demand.

Footnotes

1. This paper extends the analytic sequence developed in the preceding papers. Paper 1 separated visible usage from retained demand; Paper 2 distinguished infrastructure expansion from settlement closure and residual overflow from selection; Paper 3 translated retained demand into required liquidity-density conditions. The present paper moves from those framework conditions to applied stack-outcome analysis.
2. Ripple- and XRP-adjacent infrastructure is used in this paper as an applied stress environment rather than as proof of XRP necessity. The paper does not claim that Ripple's institutional expansion, RLUSD adoption, XRPL activity, or related infrastructure development automatically implies XRP-specific retained demand.

3. This paper is a methodological working paper. It does not provide investment advice, a price target, or a recommendation to buy, sell, or hold any digital asset.
4. Unresolved watch status is not a fourth stack outcome. It is a temporary audit state used where the available evidence is insufficient to classify a layer as compression, bypass, or amplification.

Keywords

Digital Assets; Retained Demand; Liquidity Density; Stack Audit; Compression; Bypass; Amplification; Institutional Settlement; Stablecoins; Tokenized Assets; XRP; Ripple; Collateral; Removal Sensitivity; Market Structure

Author Note

This paper is presented as a working paper and should be read as a framework-oriented contribution. It is the fourth paper in a retained-demand audit series for institutionally connected digital assets. Its purpose is to clarify how institutional stack expansion may be evaluated when asset-specific retained demand, required liquidity-density burden, compression, bypass, and amplification are treated as analytically distinct questions. Ripple- and XRP-adjacent infrastructure is used as an applied stress environment rather than as proof of XRP necessity. The contribution of the paper is methodological rather than advocative. The paper does not provide investment advice, a price target, or a recommendation to buy, sell, or hold any digital asset.

Short Description

A working paper proposing an applied stack-audit framework for institutionally connected digital assets. The paper examines whether institutional infrastructure expansion compresses, bypasses, or amplifies asset-specific retained demand and required liquidity-density burden, and uses a removal test to distinguish ecosystem growth from operational necessity. Ripple- and XRP-adjacent infrastructure is used as an applied stress environment rather than as proof of XRP necessity.

References

Basel Committee on Banking Supervision. 2013. Monitoring Tools for Intraday Liquidity Management. Bank for International Settlements.

Committee on Payments and Market Infrastructures. 2014. Developments in Collateral Management Services. Bank for International Settlements.

Committee on Payments and Market Infrastructures. 2023. A Glossary of Terms Used in Payments and Settlement Systems. Bank for International Settlements.

Financial Stability Board. 2024. Liquidity Preparedness for Margin and Collateral Calls. Financial Stability Board.

Meta, S. 2026. From Usage to Retained Demand: A Structural Audit Framework for Institutionally Connected Digital Assets. Working Paper.

Meta, S. 2026. Expansion Is Not Closure: Settlement Stack Competition and the Conditional Relevance of External Connective Assets. Working Paper.

Meta, S. 2026. From Retained Demand to Required Liquidity Density: A Conditional Sizing Framework for Institutionally Connected Digital Assets. Working Paper.
Testi, Stefano, et al. 2023. Intraday Liquidity around the World. BIS Working Papers No. 1089. Bank for International Settlements.